# Experiment (5)
# MOSFET IV Characteristics

## Objectives

- To study the IV curve of a MOSFET.
- To experimentally evaluate the threshold voltage of a MOSFET.
- To graphically find the operating point of a MOSFET circuit.

## Pre-lab Work

Analyze the DC circuit below. Assume the transistor is in saturation and find expressions for the drain current and the drain-source voltage. Ignore channel length modulation.

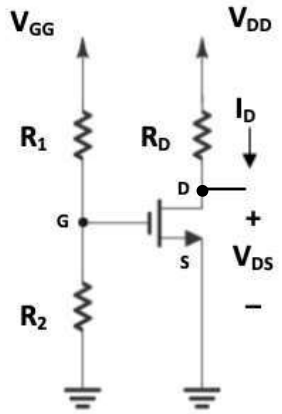



Figure 1: A MOSFET DC Circuit.

Solution

## Equipment

Table 1. Equipment for experiment 5.

| | Equipment | Model | Serial No. | Stock No. |
|---|---|---|---|---|
| 1 | BreadBoard Station | | | |
| 2 | Resistors 1 kΩ (1), 1 MΩ (2) | | | |
| 3 | NMOS transistor (1) | 2N7000/7002 | http://pdf.datasheetcatalog.com/datasheet/calogic/2N4351.PDF | |
| 4 | Multi-meter (1) | | | |
| 5 | Probes and cables | | | |

## Experimental Procedure

Part (1): MOSFET IV Characteristics.

1. Connect the circuit in figure 1 with $R_D$=1 kΩ and $R_1$= $R_2$=1 MΩ. Pay a special attention to how you connect the terminals of the MOSFET. Refer to the data sheet to identify the

transistor terminals. Make sure the Body terminal (if exists) is connected to the source terminal.

2. Set $V_{GG}$ to a constant voltage of +8 V while connecting $V_{DD}$ to a variable DC voltage source.
3. The IV characteristics ($I_D$ versus $V_{DS}$) is obtained by slowly varying $V_{DD}$ (from 0 to +10 V) and measuring the voltage $V_{DS}$ and $V_{R_D}$ at each setting. Use table 2 to enter your results.
4. Repeat step 3 for another value of $V_{GG}$, say +10 V. Record your results in table 2.

Part (2): Graphical Analysis of a DC MOSFET Circuit.

1. Reconsider the circuit in figure 1.
2. The load line equation for this circuit is $I_D = \frac{V_{DD}-V_{DS}}{R_D}$. It is the line connecting the two points $(0,\frac{V_{DD}}{R_D})$ and $(V_{DD},0)$ on the IV curve.

Part (3): MOSFET Threshold Voltage.

Produce a procedure to find the threshold voltage of the transistor using the circuit in figure 1. Write the steps below and implement them.

1. First we have to set Voltage VGG equal to zero.

2. Then we have to increase the voltage VGG step by step to reach a value equal to the Voltage VDD.

3. We measure the current across the RD at each step.

4. We plot the Id vs VGG graph. We will get an exponential graph where the rise point Id will give us the threshold voltage.

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**Results and Analysis**

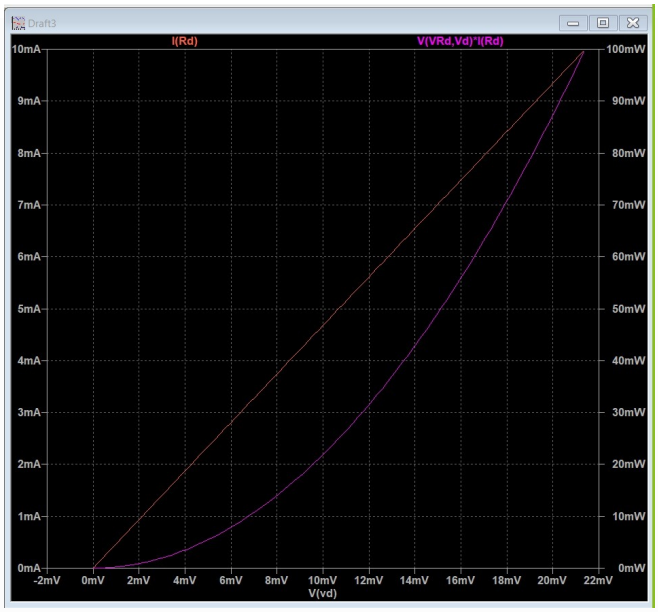
Part (1): MOSFET IV Characteristics.

1. Calculate $V_{GS}$ and the current $I_D$ values in table 2, for both values of $V_{GG}$.
2. Use graph 1 to sketch the IV curves for the two $V_{GS}$ values.
3. Comment on the IV curves that you obtained. Can you identify the transistor modes of operation?

Answer: 3-Increasing the VGG voltage from 8V to 10V did not change the IV curve characteristic. We obtained the same curve in both cases. The IV curve we obtained is linear. For this reason, we can say that the transistor operates in the ohmic region.

Table 2. IV Characteristics Measurements.

| | $V_{GG}$ = 8V, $V_{GS}$ =4V | | | $V_{GG}$ = 10V, $V_{GS}$ =5V | | |
|---|---|---|---|---|---|---|
| $V_{DD}$ (V) | $V_{DS}$ (V) | $V_{R_D}$ (V) | $I_D$ (mA) | $V_{DS}$ (V) | $V_{R_D}$ (V) | $I_D$ (mA) |
| 0.0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1.0 | 2.705mV | 0.997V | 0.997mA | 2.130mV | 0.997V | 0.997mA |
| 2.0 | 5.413mV | 1.994V | 1.994mA | 4.261mV | 1.995V | 1.995mA |
| 3.0 | 8.124mV | 2.991V | 2.991mA | 6.395mV | 2.993V | 2.993mA |
| 4.0 | 10.839mV | 3.989V | 3.989mA | 8.528mV | 3.991V | 3.991mA |
| 5.0 | 13.557mV | 4.986V | 4.986mA | 10.664mV | 4.989V | 4.989mA |
| 6.0 | 16.278mV | 5.983V | 5.983mA | 12.800mV | 5.987V | 5.987mA |
| 7.0 | 19.003mV | 6.981V | 6.981mA | 14.938mV | 6.985V | 6.985mA |
| 8.0 | 21.731mV | 7.978V | 7.978mA | 17.078mV | 7.982V | 7.982mA |
| 9.0 | 24.462mV | 8.975V | 8.975mA | 19.219mV | 8.980V | 8.980mA |
| 10.0 | 27.197mV | 9.972V | 9.972mA | 21.361mV | 9.978V | 9.978mA |




Graph 1: MOSFET IV Curves.

4. Is the drain current constant in the saturation region of the IV curve? Explain why or why not.

Answer: The drain current in the saturation region of the IV curve is a constant under normal conditions. However, since our circuit operates in the ohmhic region, we cannot comment on the behavior of the drain current in the saturation region.

Part (2): Graphical Analysis of a DC MOS Circuit.

1. Draw the load line of the circuit in figure 1 with $V_{DD}$ and $V_{GG}$ both connected to +10 V. Draw the line on top of the IV curves in graph 1.
2. Read the drain current value and the drain-source voltage ($V_{DS}$) value from the intersection point in graph 1. Record the results in table 3.
3. Use the results in table 2 for the experimental $V_{DS}$ and $I_D$.

Table 3. Graphical Solution Results.

| $V_{DD}$= 10 V and $V_{GG}$ = 10 V | | | | | |
|---|---|---|---|---|---|
| $V_{DS}$ (V) Experimental value | $V_{DS}$ (V) Graphical value | % Difference | $I_D$ (mA) Experimental value | $I_D$ (mA) Graphical value | % Difference |
| 21.361mV | 21.361mV | 0 | 9.978mA | 9.978mA | 0 |

4. Comment on your results explaining the sources of discrepancy.

Answers:

4- Since we conducted the entire experiment on LTspice, we cannot comment on the sources of inconsistency because our experimental and graphical results are very close to each other.

Part (3): MOSFET Threshold Voltage.

1. Comment on your result regarding the threshold voltage as compared to the data sheet value.
2. Comment on the procedure that you used to find the threshold voltage. How can it be improved?
3. Can you estimate the value of the constant $k_n$ for this transistor? Show your work.

Answers: 1- When we compare it to the data sheet value, we observed that the threshold voltage is close to correct.

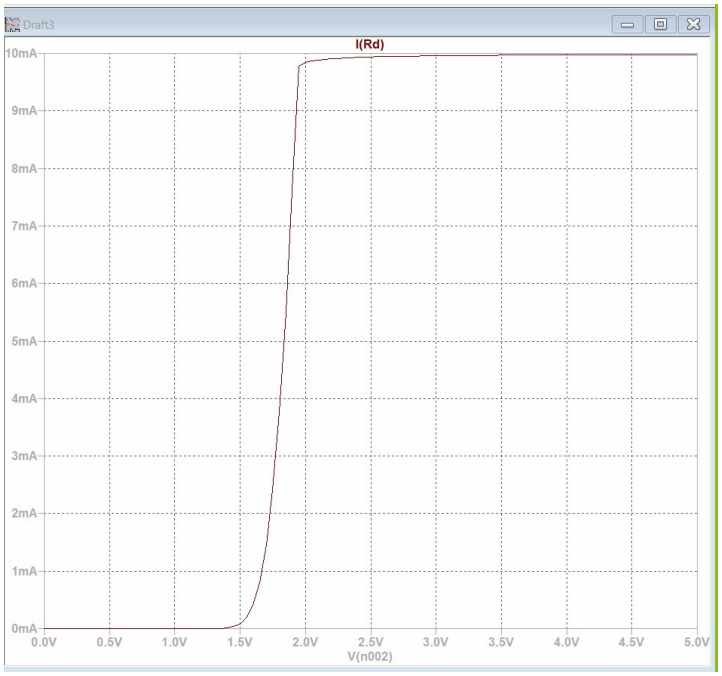
2- The procedure you used to find the threshold voltage was sufficient, we can use it the same way without any improvement.
Because the accuracy of the values we get is quite high.

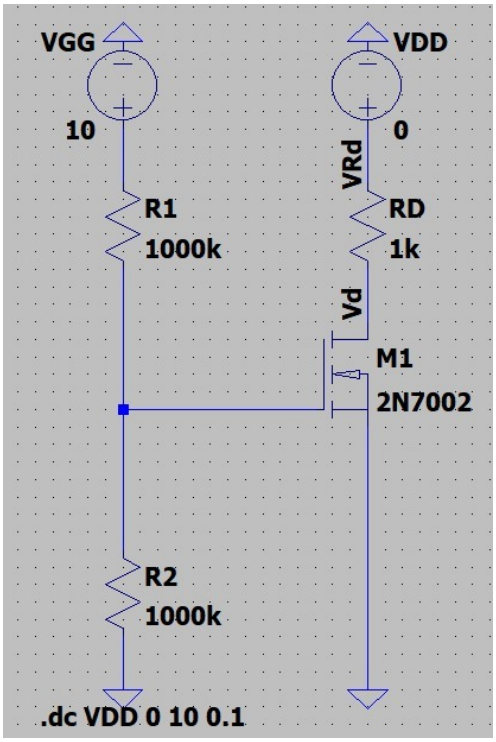
## LTspice Simulations

Include with your report LTspice simulation results of all parts of this experiment.

## Conclusions and Remarks

Write below all your conclusions/remarks on this experiment.




| Name | Nurullah Said Tüğmen | Name | İlayda Kaya | Name | | Name | |
|---|---|---|---|---|---|---|---|
| ID | 150720823 | ID | 150720502 | ID | | ID | |
| Section | 1 | | | | | | |

# N-Channel 60-V (D-S) MOSFET

| PRODUCT SUMMARY | | | | |
|---|---|---|---|---|
| **Part Number** | **$V_{(BR)DSS}$ Min (V)** | **$r_{DS(on)}$ Max (Ω)** | **$V_{GS(th)}$ (V)** | **$I_D$ (A)** |
| 2N7000 | 60 | 5 @ $V_{GS}$ = 10 V | 0.8 to 3 | 0.2 |
| 2N7002 | | 7.5 @ $V_{GS}$ = 10 V | 1 to 2.5 | 0.115 |
| VQ1000J | | 5.5 @ $V_{GS}$ = 10 V | 0.8 to 2.5 | 0.225 |
| VQ1000P | | 5.5 @ $V_{GS}$ = 10 V | 0.8 to 2.5 | 0.225 |
| BS170 | | 5 @ $V_{GS}$ = 10 V | 0.8 to 3 | 0.5 |

## FEATURES

- Low On-Resistance: 2.5 Ω
- Low Threshold: 2.1 V
- Low Input Capacitance: 22 pF
- Fast Switching Speed: 7 ns
- Low Input and Output Leakage

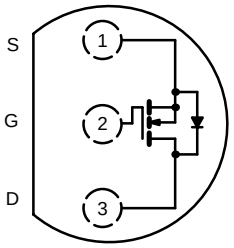
## BENEFITS

- Low Offset Voltage
- Low-Voltage Operation
- Easily Driven Without Buffer
- High-Speed Circuits
- Low Error Voltage

## APPLICATIONS

- Direct Logic-Level Interface: TTL/CMOS
- Drivers: Relays, Solenoids, Lamps, Hammers, Displays, Memories, Transistors, etc.
- Battery Operated Systems
- Solid-State Relays

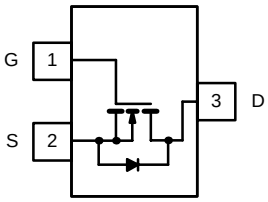
**TO-226AA**
**(TO-92)**




Top View

2N7000

**TO-236**
**(SOT-23)**

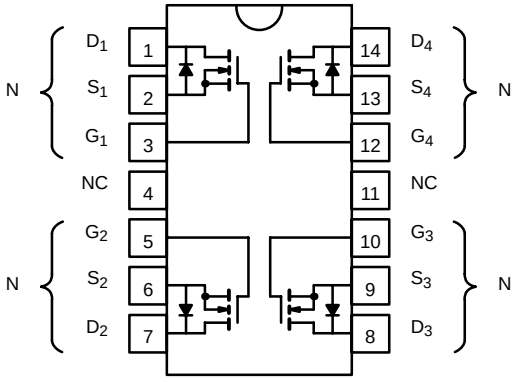



Top View

Marking Code: 72*wll*

72 = Part Number Code for 2N7002
*w* = Week Code
*ll* = Lot Traceability

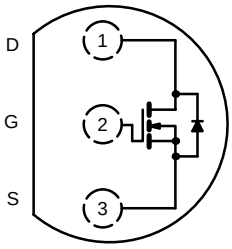
**Dual-In-Line**




Top View

Plastic: VQ1000J
Sidebraze: VQ1000P

**TO-92-18RM**
**(TO-18 Lead Form)**




Top View

BS170

## ABSOLUTE MAXIMUM RATINGS ($T_A$ = 25°C UNLESS OTHERWISE NOTED)

| Parameter | | Symbol | 2N7000 | 2N7002 | Single VQ1000J | Single VQ1000P | Total Quad VQ1000J/P | BS170 | Unit |
|---|---|---|---|---|---|---|---|---|---|
| Drain-Source Voltage | | $V_{DS}$ | 60 | 60 | 60 | 60 | | 60 | V |
| Gate-Source Voltage—Non-Repetitive | | $V_{GSM}$ | ±40 | ±40 | ±30 | | | ±25 | V |
| Gate-Source Voltage—Continuous | | $V_{GS}$ | ±20 | ±20 | ±20 | ±20 | | ±20 | V |
| Continuous Drain Current ($T_J$ = 150°C) | $T_A$= 25°C | $I_D$ | 0.2 | 0.115 | 0.225 | 0.225 | | 0.5 | A |
| | $T_A$= 100°C | | 0.13 | 0.073 | 0.14 | 0.14 | | 0.175 | A |
| Pulsed Drain Current[a] | | $I_{DM}$ | 0.5 | 0.8 | 1 | 1 | | | A |
| Power Dissipation | $T_A$= 25°C | $P_D$ | 0.4 | 0.2 | 1.3 | 1.3 | 2 | 0.83 | W |
| | $T_A$= 100°C | | 0.16 | 0.08 | 0.52 | 0.52 | 0.8 | | W |
| Thermal Resistance, Junction-to-Ambient | | $R_{thJA}$ | 312.5 | 625 | 96 | 96 | 62.5 | 156 | °C/W |
| Operating Junction and Storage Temperature Range | | $T_J$, $T_{stg}$ | –55 to 150 | | | | | | °C |

Notes
a. Pulse width limited by maximum junction temperature.
b. $t_p \leq 50$ μs.

## SPECIFICATIONS—2N7000 AND 2N7002 ($T_A$ = 25°C UNLESS OTHERWISE NOTED)

| Parameter | Symbol | Test Conditions | Typ[a] | Limits 2N7000 Min | Limits 2N7000 Max | Limits 2N7002 Min | Limits 2N7002 Max | Unit |
|---|---|---|---|---|---|---|---|---|
| **Static** | | | | | | | | |
| Drain-Source Breakdown Voltage | $V_{(BR)DSS}$ | $V_{GS}$ = 0 V, $I_D$ = 10 μA | 70 | 60 | | 60 | | V |
| Gate-Threshold Voltage | $V_{GS(th)}$ | $V_{DS}$ = $V_{GS}$, $I_D$ = 1 mA | 2.1 | 0.8 | 3 | | | V |
| | | $V_{DS}$ = $V_{GS}$, $I_D$ = 0.25 mA | 2.0 | | | 1 | 2.5 | V |
| Gate-Body Leakage | $I_{GSS}$ | $V_{DS}$ = 0 V, $V_{GS}$ = ±15 V | | | ±10 | | | nA |
| | | $V_{DS}$ = 0 V, $V_{GS}$ = ±20 V | | | | | ±100 | nA |
| Zero Gate Voltage Drain Current | $I_{DSS}$ | $V_{DS}$ = 48 V, $V_{GS}$ = 0 V | | | 1 | | | μA |
| | | $T_C$ = 125°C | | | 1000 | | | μA |
| | | $V_{DS}$ = 60 V, $V_{GS}$ = 0 V | | | | | 1 | μA |
| | | $T_C$ = 125°C | | | | | 500 | μA |
| On-State Drain Current[b] | $I_{D(on)}$ | $V_{DS}$ = 10 V, $V_{GS}$ = 4.5 V | 0.35 | 0.075 | | | | A |
| | | $V_{DS}$ = 7.5 V, $V_{GS}$ = 10 V | 1 | | | 0.5 | | A |
| Drain-Source On-Resistance[b] | $r_{DS(on)}$ | $V_{GS}$ = 4.5 V, $I_D$ = 0.075 A | 4.5 | | 5.3 | | | Ω |
| | | $V_{GS}$ = 5 V, $I_D$ = 0.05 A | 3.2 | | | | 7.5 | Ω |
| | | $T_C$ = 125°C | 5.8 | | | | 13.5 | Ω |
| | | $V_{GS}$ = 10 V, $I_D$ = 0.5 A | 2.4 | | 5 | | 7.5 | Ω |
| | | $T_J$ = 125°C | 4.4 | | 9 | | 13.5 | Ω |
| Forward Transconductance[b] | $g_{fs}$ | $V_{DS}$ = 10 V, $I_D$ = 0.2 A | | 100 | | 80 | | mS |
| Common Source Output Conductance[b] | $g_{os}$ | $V_{DS}$ = 5 V, $I_D$ = 0.05 A | 0.5 | | | | | mS |
| **Dynamic** | | | | | | | | |
| Input Capacitance | $C_{iss}$ | $V_{DS}$ = 25 V, $V_{GS}$ = 0 V f = 1 MHz | 22 | | 60 | | 50 | pF |
| Output Capacitance | $C_{oss}$ | | 11 | | 25 | | 25 | pF |
| Reverse Transfer Capacitance | $C_{rss}$ | | 2 | | 5 | | 5 | pF |

## SPECIFICATIONS—2N7000 AND 2N7002 ($T_A$ = 25°C UNLESS OTHERWISE NOTED)

| Parameter | Symbol | Test Conditions | Typ[a] | Limits 2N7000 Min | Limits 2N7000 Max | Limits 2N7002 Min | Limits 2N7002 Max | Unit |
|---|---|---|---|---|---|---|---|---|
| **Switching[d]** | | | | | | | | |
| Turn-On Time | $t_{ON}$ | $V_{DD}$ = 15 V, $R_L$ = 25 Ω<br>$I_D$ ≅ 0.5 A, $V_{GEN}$ = 10 V, $R_G$ = 25 Ω | 7 | | 10 | | | ns |
| Turn-Off Time | $t_{OFF}$ | | 7 | | 10 | | | |
| Turn-On Time | $t_{ON}$ | $V_{DD}$ = 30 V, $R_L$ = 150 Ω<br>$I_D$ ≅ 0.2 A, $V_{GEN}$ = 10 V, $R_G$ = 25 Ω | 7 | | | | 20 | |
| Turn-Off Time | $t_{OFF}$ | | 11 | | | | 20 | |

## SPECIFICATIONS—VQ1000J/P AND BS170 ($T_A$ = 25°C UNLESS OTHERWISE NOTED)

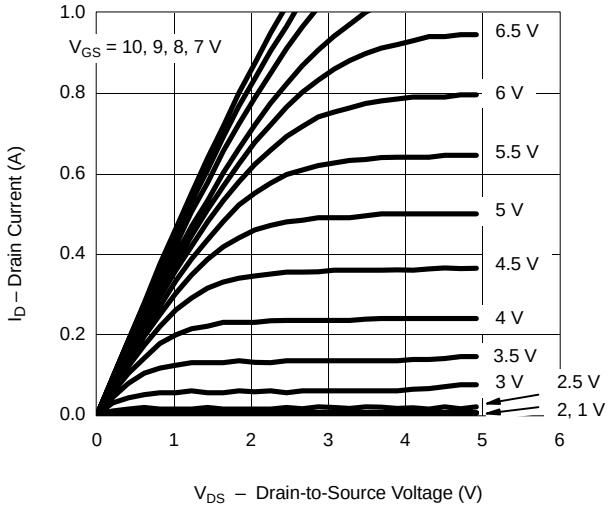
| Parameter | Symbol | Test Conditions | Typ[a] | Limits VQ1000J/P Min | Limits VQ1000J/P Max | Limits BS170 Min | Limits BS170 Max | Unit |
|---|---|---|---|---|---|---|---|---|
| **Static** | | | | | | | | |
| Drain-Source Breakdown Voltage | $V_{(BR)DSS}$ | $V_{GS}$ = 0 V, $I_D$ = 100 μA | 70 | 60 | | 60 | | V |
| Gate-Threshold Voltage | $V_{GS(th)}$ | $V_{DS}$ = $V_{GS}$, $I_D$ = 1 mA | 2.1 | 0.8 | 2.5 | 0.8 | 3 | |
| Gate-Body Leakage | $I_{GSS}$ | $V_{DS}$ = 0 V, $V_{GS}$ = ±10 V | | | ±100 | | | nA |
| | | $T_J$ = 125°C | | | ±500 | | | |
| | | $V_{DS}$ = 0 V, $V_{GS}$ = ±15 V | | | | | ±10 | |
| Zero Gate Voltage Drain Current | $I_{DSS}$ | $V_{DS}$ = 25 V, $V_{GS}$ = 0 V | | | | | 0.5 | μA |
| | | $V_{DS}$ = 48 V, $V_{GS}$ = 0 V, $T_J$ = 125°C | | | 500 | | | |
| | | $V_{DS}$ = 60 V, $V_{GS}$ = 0 V | | | 10 | | | |
| On-State Drain Current[b] | $I_{D(on)}$ | $V_{DS}$ = 10 V, $V_{GS}$ = 10 V | 1 | 0.5 | | | | A |
| Drain-Source On-Resistance[b] | $r_{DS(on)}$ | $V_{GS}$ = 5 V, $I_D$ = 0.2 A | 4 | | 7.5 | | | Ω |
| | | $V_{GS}$ = 10 V, $I_D$ = 0.2 A | 2.3 | | | | 5 | |
| | | $V_{GS}$ = 10 V, $I_D$ = 0.3 A | 2.3 | | 5.5 | | | |
| | | $T_J$ = 125°C | 4.2 | | 7.6 | | | |
| Forward Transconductance[b] | $g_{fs}$ | $V_{DS}$ = 10 V, $I_D$ = 0.2 A | | | | 100 | | mS |
| | | $V_{DS}$ = 10 V, $I_D$ = 0.5 A | | 100 | | | | |
| Common Source Output Conductance[b] | $g_{os}$ | $V_{DS}$ =5 V, $I_D$ = 0.05 A | 0.5 | | | | | |
| **Dynamic** | | | | | | | | |
| Input Capacitance | $C_{iss}$ | $V_{DS}$ =25 V, $V_{GS}$ = 0 V<br>f = 1 MHz | 22 | | 60 | | 60 | pF |
| Output Capacitance | $C_{oss}$ | | 11 | | 25 | | | |
| Reverse Transfer Capacitance | $C_{rss}$ | | 2 | | 5 | | | |
| **Switching[d]** | | | | | | | | |
| Turn-On Time | $t_{ON}$ | $V_{DD}$ = 15 V, $R_L$ = 23 Ω<br>$I_D$ ≅ 0.6 A, $V_{GEN}$ = 10 V, $R_G$ = 25 Ω | 7 | | 10 | | | ns |
| Turn-Off Time | $t_{OFF}$ | | 7 | | 10 | | | |
| Turn-On Time | $t_{ON}$ | $V_{DD}$ = 25 V, $R_L$ = 125 Ω<br>$I_D$ ≅ 0.2 A, $V_{GEN}$ = 10 V, $R_G$ = 25 Ω | 7 | | | | 10 | |
| Turn-Off Time | $t_{OFF}$ | | 7 | | | | 10 | |

Notes
a. For DESIGN AID ONLY, not subject to production testing.
b. Pulse test: PW ≤ 80 μs duty cycle ≤ 1%.
c. This parameter not registered with JEDEC.
d. Switching time is essentially independent of operating temperature.

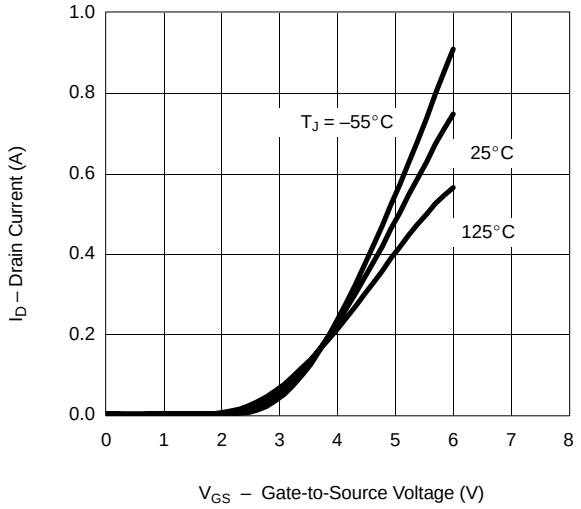
VNBF06

## TYPICAL CHARACTERISTICS ($T_A$ = 25°C UNLESS OTHERWISE NOTED)

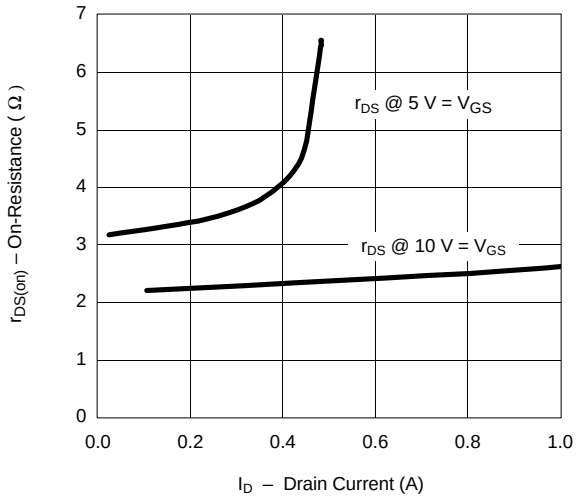
**Output Characteristics**



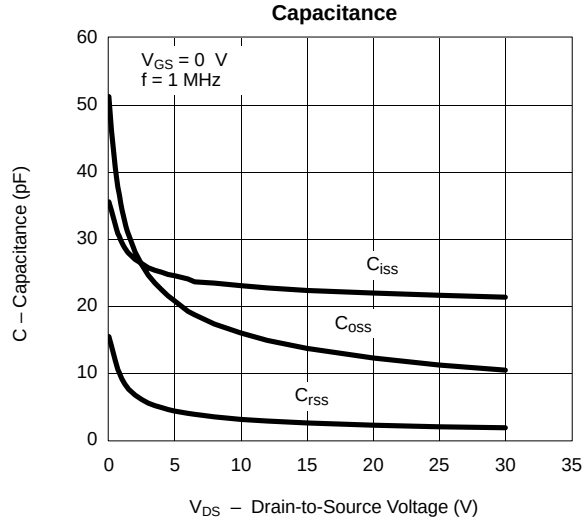

**Transfer Characteristics**



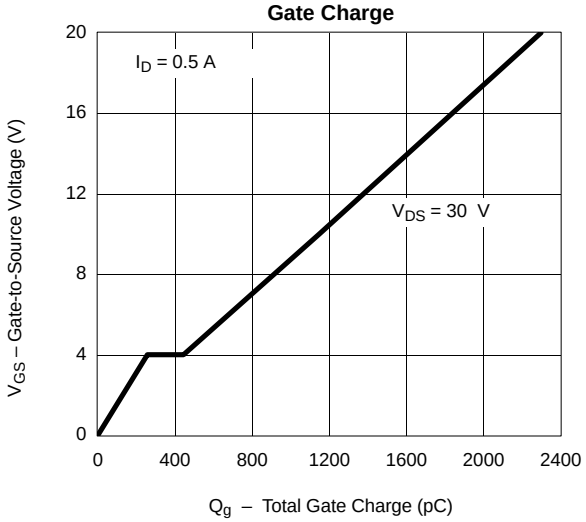

**On-Resistance vs. Drain Current**



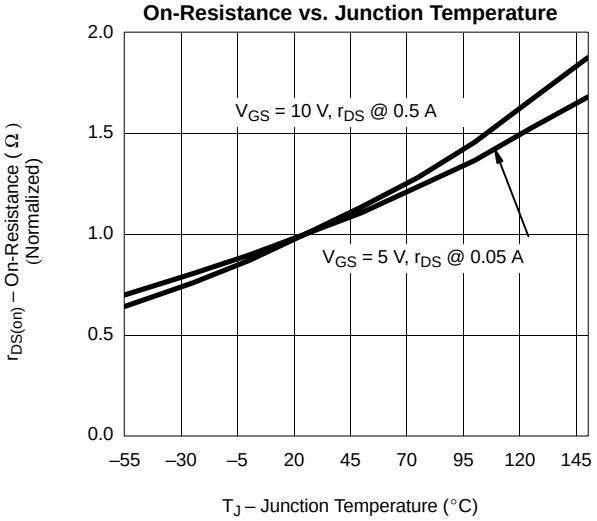

**Capacitance**




**Gate Charge**

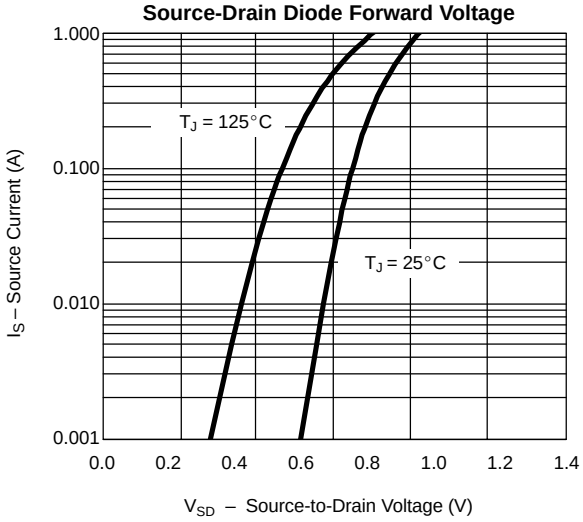



**On-Resistance vs. Junction Temperature**

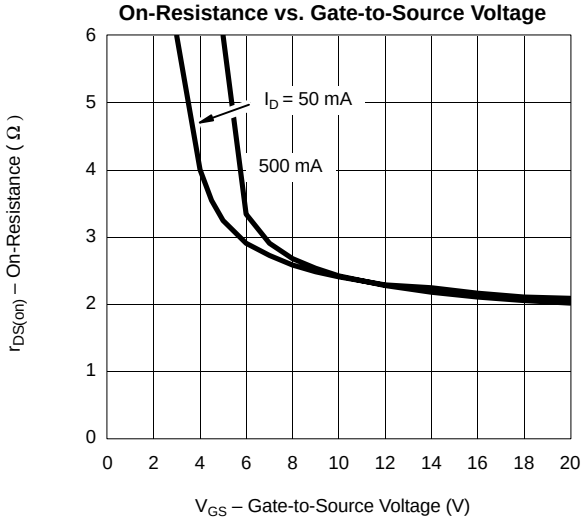


## TYPICAL CHARACTERISTICS ($T_A$ = 25°C UNLESS OTHERWISE NOTED)

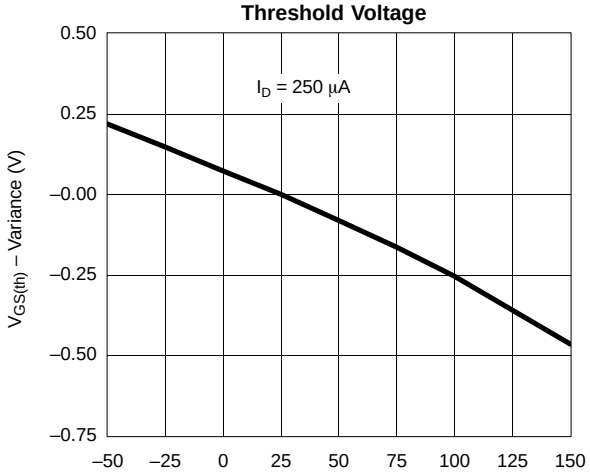
**Source-Drain Diode Forward Voltage**



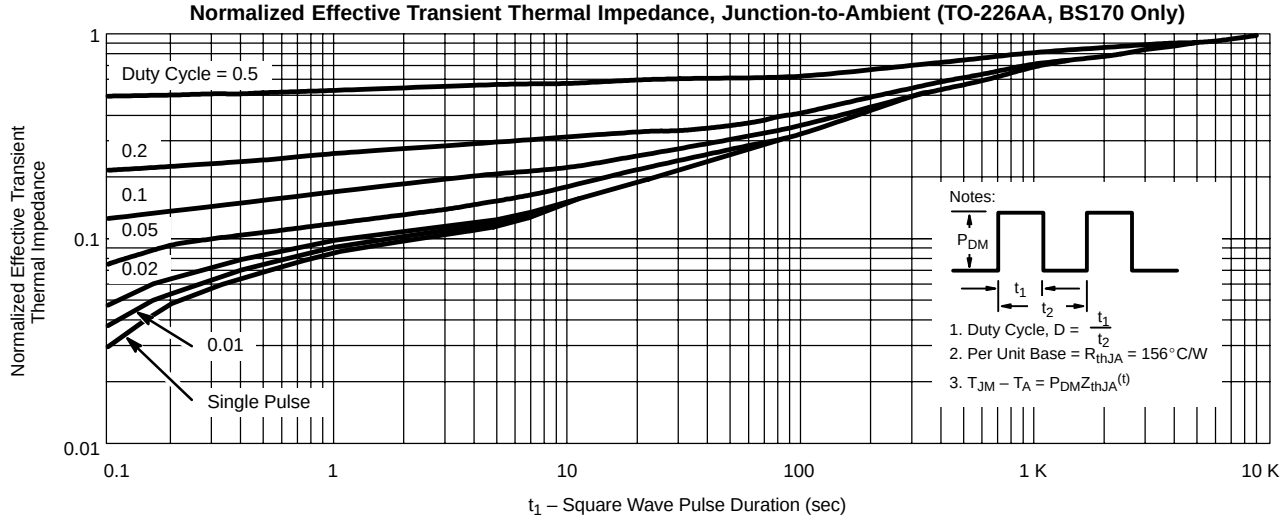

**On-Resistance vs. Gate-to-Source Voltage**




**Threshold Voltage**




**Normalized Effective Transient Thermal Impedance, Junction-to-Ambient (TO-226AA, BS170 Only)**



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